

step-down transformer (X1), two 1N4007 rectifier diodes (D1 and D2) and a 5V regulator 7805 (IC1).

The cube uses 64 green LEDs to make a 4 rows $\times$ 16 columns structure. All the LEDs are wired to

74HC154 decoder and AT89C2051 microcontroller. AT89C2051 is a 20-pin MCS-51 family controller having only two input/output ports.

To control individual LEDs for different display patterns, the 4 rows

(first layer) down towards the base layer (fourth layer) on the PCB. Take a template, preferably a cardboard with holes as shown in Fig. 3(a). Hole-to-hole distance is about 25.4mm. The cardboard helps in holding the LEDs during lead soldering.

Insert 16 LEDs into the holes and bend the negative (cathode/shorter) leads 90° outwards using needle-nose pliers. Solder negative leads of all the LEDs in the first layer. Next, solder positive leads of all the LEDs in the second layer as shown in Fig. 3(b). Connect the subsequent layers in this manner. Solder positive leads of all the LEDs in the fourth layer to the PCB board.

The final structure of the 4x4x4 LED cube is shown in Fig. 4.

Software

The circuit is controlled using the software program loaded into the internal 2kB memory of AT89C2051. The program implements all the required functionalities. It is written in embedded C programming language and compiled using Keil μ vision 4 IDE.

The code has five loops (stage1,

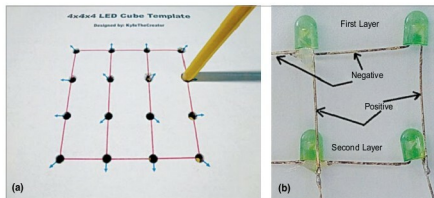


Fig. 3: Tips for soldering: (a) Placement of LEDs on the template, and (b) Bending/pulling the negative lead

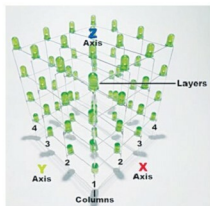


Fig. 4: Final structure of the 4x4x4 LED cube

are controlled by BC547 transistors (T1 through T4). Port pins P1.0 through P1.3 control row transistors T4 through T1, respectively, and pins P1.4 through P1.7 control decoder IC3 with the help of the software program.

First, using a button cell battery or multimeter in diode mode, check each LED for proper working. This may sound tedious but, in the end, it will safeguard your project!

Cube construction. Construct LED layers from the topmost layer

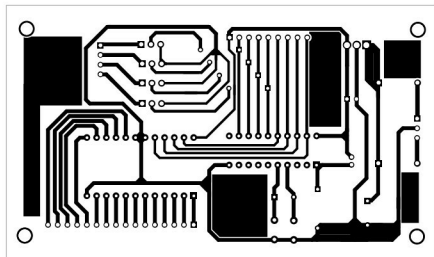


Fig. 5: Actual-size PCB layout of the 4x4x4 LED cube

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - AT89C2051 microcontroller
- IC3 - 74HC154 decoder
- T1-T4 - BC547 npn transistor
- D1, D2 - 1N4007 rectifier diode
- LED1-LED64 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 10-kilo-ohm
- R5 - 8.2-kilo-ohm
- RNW1 - 10-kilo-ohm resistor network

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 100 μ F, 16V electrolytic
- C3 - 10 μ F, 16V electrolytic
- C4, C5 - 33pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 500mA secondary transformer
- CON1 - 3-pin connector
- CON2 - 16-pin connector
- CON3 - 4-pin connector
- XTAL1 - 11.0592MHz
- 230V AC mains
- 20-pin IC socket
- 24-pin IC socket
- Head-sink for IC1